



Q-quatics
ANNUAL REPORT
2023

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Overview

The year 2023 was marked by a pivoting change for Quantitative Aquatics (Q-quatics) in terms of fully transitioning from staff working remotely to revitalizing our office operations with the presence of all employees. The scourge of the global pandemic has now been replaced by vibrant activities throughout the year.

Our usual and committed partners (University of British Columbia UBC, Helmholtz Centre for Ocean Research Kiel GEOMAR, The Swire Institute of Marine Science SWIMS, University of Western Australia UWA, WorldFish, Muséum national d'Histoire naturelle MNHN, Swedish Museum of Natural History NRM) led our project-funded work while we engaged with new partners (Ministry of Agriculture, Fisheries, Wealth and Water Resources MAFWR, University of Exeter UoE, University of Lorraine UL, Perry Institute of Marine Science PIMS, Monterey Bay Aquarium MBA) to form new collaborations where some extended beyond 2023.

Members of the FishBase Consortium displayed a significant presence in this year's work plan through five funded projects with Q-quatics.

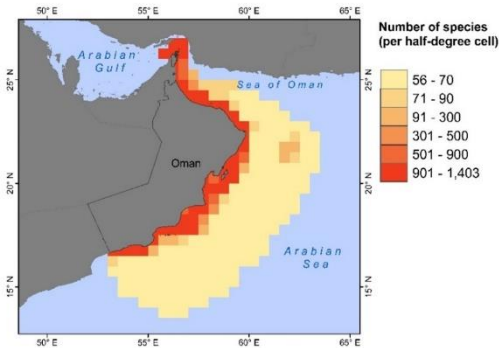
The Board of Trustees initiated planning for a fund-raising activity with personalities engaged in environment, conservation, and education in the Philippines, including business owners, philanthropists, and politicians.

As we are successfully expanding collaborations, we were also faced with the challenge of searching for another person to lead Q-quatics as our Executive Director has reached her retirement age.

Delivering Impact

Project Highlights

With the *Sea Around Us*, a project for the Sultanate of Oman was completed which aimed to update and enhance the country's marine biodiversity data in FishBase and SeaLifeBase; reconstruct marine fisheries catches within their Exclusive Economic Zone (EEZ); and provide a more sensitive Regional Marine Trophic Index (RMTI) that performs without the strong effects of low-trophic level organisms. The product of this project



was instrumental in improving the country's ranking in the global Environmental Performance Index (EPI) and was duly acknowledged and lauded by the members of the Ministry of Agriculture, Fisheries, Wealth and Water Resources (MAFWR), Sultanate of Oman.

Through collaboration with the Swire Institute of Marine Science (SWIMS), Q-quatics teamed up with experts from the University of Hong Kong (HKU) to develop a research tool that analyses genetic diversity of species within Hong Kong and its adjacent regions. This project runs for three years ending in 2025 and aims to develop a Genetic Diversity Risk Indicator that will pool genetic information of species important to fisheries and aquaculture in one platform to assess the genetic diversity level of species in Hong Kong and surrounding regions as a contributing means to focus conservation strategies and preservation of fish species by relevant stakeholders in the

country and elsewhere. A well-performing tool will be of interest to other regions too.

With the Aristotle University of Thessaloniki (AUTH), another research project kicked-off in 2023 under the EcoScope initiative funded by the Horizon Europe Program. EcoScope stands for ‘Ecocentric management for sustainable fisheries and healthy marine ecosystems’. Q-quatics collaborated in several work packages documenting information on marine species as inputs to AquaMaps. The aim is to aid in predicting and influencing the direction of mitigation advice provided for fisheries strategic planning related to climate change responses. Q-quatics partners with 23 universities and research institutions around the world for this project.

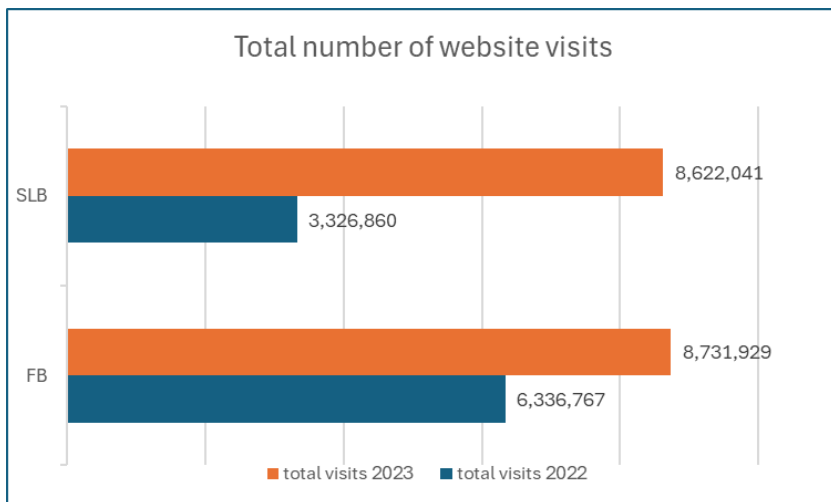


Usage of FishBase and SeaLifeBase

Website users of FishBase and SeaLifeBase in terms of number of visits, unique users, and collaborators increased from 2022 levels by 27%, 24% and 1% (for FishBase) and 61%, 32% and 3% (for SeaLifeBase), respectively. This represents the highest usage increase of the SeaLifeBase website since it was made available online in 2008 after its launch in 2005.

Table 1. FishBase and SeaLifeBase usage, and number of collaborators, 2023.

	FishBase		SeaLifeBase	
	Total	Ave/Mo	Total	Ave/Mo
Visits	8,731,929	727,660	8,622,041	718,503
Unique users	4,038,119	336,509	806,583	67,215
Total collaborators	2,175		361	



Feedback about the merits of FishBase and SeaLifeBase via email and social media underscores the reputation among users in research, policy, and advocacy for the conservation and protection of aquatic resources. They value these scientifically validated open access sources for their work linked to the care for the environment. Appendix 1 compiles the feedback received in 2023. This resonates with the findings of the study conducted by Humphries et al. 2023 (doi.org/10.26028/cybium/2023-002) on the scientific impact of FishBase after three decades. The study illuminates the rising global importance of FishBase and the diversity of disciplines that use it (see below, Publications).

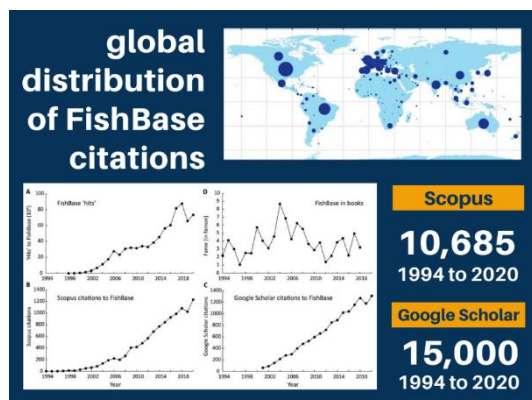
Publications

The Q-quatics list of publications for 2023 comprises of 45 papers written and published by our staff, partners and collaborators. See the full List of Publications in Appendix 2. Many of these papers were only possible, thanks to the global coverage and influence of FishBase and SeaLifeBase. We identify a couple of particularly interesting research findings:

Froese R, Pauly D (2023) "Metabolic scaling is the product of life-history optimization". Science 380(6643):eade6084.

According to Froese (GEOMAR) and Pauly (The Sea Around Us), both responsible for the launch of FishBase more than 30 years ago, the growth of fish doesn't slow down when and because they start spawning. "Fish do not have to 'choose' between growth or reproduction because, in the real world, these do not occur simultaneously but rather sequentially," says Pauly. In fact, the authors claim fish growth accelerates after they reproduce, supporting it with the concept of the Gill-Oxygen Limitation Theory (GOLT).

Humphries AT, Dimarchopoulou D, Stergiou KI, Tsikliras AC, Palomares MLD, Bailly N, Nauen CE, Luna S, Banasihan LP, Froese R, Pauly D (2023) Measuring the impact of FishBase after three decades. Cybium 47(3):213-224.



Results of the citation analysis for the period 1994 to 2020 show that FishBase is one of the most highly cited public resources (top 1%) in both the current and previous centuries. Findings also show that FishBase is most widely used in

Europe, United States of America, Brazil, and Australia from top authors primarily in agricultural, biological and environmental sciences, specifically those working on fisheries biology and management, as well as parasitology, among others.

Prellezo R, Da-Rocha JM, Palomares MLD, Sumaila UR, Villasante S (2023) Building climate resilience, social sustainability and equity in global fisheries. npj Ocean Sustainability 2(1):10.

Results reveal that the monetary benefits of the market-based solutions could reach \$66 billion per year by 2050 (122 million tonnes of CO₂e sequestered per year) if the social cost of carbon dioxide (SC-CO₂), which refers to the monetized value of the damages to society caused by each incremental tonne of CO₂, is also considered. Palomares said that if CO₂ trading prices reach the 2050 SC-CO₂, around 75% of the landings worldwide would be more valuable as carbon than as foodstuff in the market. This scheme provides the global economy with a tool to ponder options for the fisheries sector which grapples with the complexity of reallocating invested capital. It also promotes a reduction in overfishing and contributes to building climate resilience and a more equitable distribution of income from the oceans. Naturally, in exploring alternative pathways it must be kept in mind that natural resources are not fungible like finance and food will still be required in the future.

Our Growing Team



Dr Marivic Pajaro joined as the sixth member of the Q-quatics Board of Trustees in October 2023. Dr Pajaro leads an NGO (Daluhay Daloy ng Buhay) based in Aurora Province, Philippines, since 2010. Her expertise is on community-based management of marine protected areas, participatory education, and local knowledge for promoting community coastal resources management, ethno-ecology, and biodiversity conservation

engaging various stakeholders, including indigenous communities.

The Q-quatics Board of Trustees (BOT) is composed of Dr Cornelia E Nauen (Chair); Dr Fabrice Teletchea (Vice-Chair), Dr Benjamin Vallejo Jr (Corporate Secretary), Ms Flor Maclean (Treasurer), and Dr Maria Lourdes Palomares (Member). See Appendix 3 for the affiliations of the Q-quatics BOT. Dr Nauen convened the Annual Meeting which was held at the Q-quatics office in Los Baños, Philippines on 28 February 2023, and a special meeting via zoom on 8 September 2023.

The Q-quatics team consisted of 25 regular staff in 2023: Research (13), Information Technology (5), Common Services (3), Administration (4) and Consultant (1). See Appendix 4 for the list of Q-quatics staff.

Several staff were recruited for new projects. Two Research Assistants were hired, one as replacement for a Research Assistant who resigned in November 2023, and another to help in forecast influx of data for georeferencing for the DigIn Project. One more project Research Assistant and a Software Engineer to assist in in-house database will come on board early 2024.



Projects and Collaborations

We started 13 new projects in 2023, of which five projects were with new partner institutions (namely the Ministry of Agriculture, Fisheries, Wealth and Water Resources MAFWR, University of Exeter UoE, University of Lorraine UL, Perry Institute of Marine Science PIMS, Monterey Bay Aquarium MBA), five projects involved members of the FishBase Consortium (FBC) (namely

Helmholtz Centre for Ocean Research Kiel GEOMAR, The Swire Institute of Marine Science SWIMS, Muséum national d'Histoire naturelle MNHN, WorldFish, Swedish Museum of Natural History NRM). Three projects were with recurring partners (*Mundus maris*, University of Minnesota UMN, Helmholtz Institute for Functional Marine Biodiversity HIFMB). Eight projects were completed in 2023, four projects will end in 2024, and six projects will run until 2027.

Most of the projects were done in collaboration with the Sea Around Us of UBC (SAU-UBC, also a member of the FBC) by virtue of the Agreement between Q-quatics and SAU-UBC to conduct research of common interest and securing funds for joint projects which include global fisheries and biodiversity conservation. Moreover, two software engineers of Q-quatics under the supervision of the SAU Manager maintain the SAU websites and the SAU reconstructed fisheries catches database along with attending to other IT needs. Q-quatics continued to serve the research and policy needs using SAU time series global cell data by fishing entity, taxon name, commercial group, year and catch. See Appendix 5 for the list of completed and on-going projects until 2027, data services and core funding, for the year 2023.

Stock Assessment Data Provider



We continued to provide fisheries stock assessments from catch data and published references that inform good practice decision-making for sustainable fisheries. Updated stock assessment spreadsheets were provided to private seafood retailers and non-governmental NGOs who use these as reference to gauge if the fish stocks are sustainably harvested.

Species Sponsorship at FishBase and SeaLifeBase

We proactively persuade power users of our databases to contribute funds to FishBase and SeaLifeBase on a regular schedule, even in small amounts. This is important to maintain our public service in the face of rising costs. We continue to offer a species sponsorship scheme that displays the logo of the sponsoring agencies on the summary page of their sponsored species. The proceeds of the program support FishBase and SeaLifeBase encoding and development. In 2023, institutional sponsorship was received from UWA with 274 species, *Mundus maris* with 21 species, and Fair Fish International and Nature Metrics Ltd with 10 species each. We have an individual sponsorship from Dr Charles Blend of the Corpus Christi Museum of Science and History who has been sponsoring one species since 2012.

Library Collections

The Q-quatics Library was the initial beneficiary of a donation of more than 500 books and 9,000 reprints of scientific papers, mostly on fish and fisheries-related topics from the personal collections of Dr Daniel Pauly. This rich reference pool was generated from his more than six decades of research and academic teaching, working in the Philippines some years in the 1980s and 1990s. Books in duplicate and the reprints once digitized, will be donated to local fisheries libraries following Dr Pauly's request. Even with advances in technology, libraries in the suburbs and remote community settlements are still in dire need of printed reference materials.

Three issues of the Q-quatics e-News were published in 2023: [Issue No. 16, April 2023](#); [Issue No. 17, August 2023](#); and [Issue No. 18, December 2023](#).

Corporate Branding

In an effort to improve the corporate branding of Q-quatics databases, [new logos of FishBase, SeaLifeBase and AquaMaps](#) were created. The new set of logos show a harmonized look so users would easily perceive the strong collaboration and complementarity of our data information systems.



FishBase Consortium

The FishBase Consortium Coordinator, Dr Maria Lourdes Palomares (who is also the Q-quatics Science Director) coordinated two FBC interim meetings in 2023: the first Quarterly Meeting on 19 January and the second Quarterly Meeting on 11 May, both held via Zoom, and hosted by SAU-UBC. These meetings discussed work progress, financial status of the encoding team in Los Baños, and external collaboration. The Annual Meeting, a physical meeting (with provision of online participation since the Covid-19 outbreak), was hosted by the Royal Museum for Central Africa (MRAC) in Tervuren, Belgium. The Annual Meeting was preceded by the Annual FishBase and SeaLifeBase Symposium. As the elected FBC Chair for 2022-2023, Dr Jos Snoeks with his FishBase team managed the two events. Dr Snoeks took over from Dr Alexander Tilley of WorldFish, 2021-2022 Chair who organized the annual general meeting events in Penang, Malaysia in 2022.

Dr Athanassios Tsikliras of the Aristotle University of Thessaloniki, Greece was elected as FBC Chair for 2023-2024. The 2024 Annual FishBase/SeaLifeBase symposium and meeting will be hosted in Thessaloniki in 2-3 September. The symposium schedule is as follows: Day 1: Advances in Gill Oxygen Limitation Theory, and Day 2: Fishes in Mare Nostrum,

and other areas. The 3-day FBC Annual Meeting (4-6 September) will host invited observers and guests from West Africa.

Our Performance

Statement of Financial Activities

The summary of revenue and expenses of Q-quatics for 2023 are presented in Table 2. Details on the project name, duration and partners are given in Appendix 5.

Table 2. Q-quatics revenue by source and expenses by type, January to December 2023.

Revenue	'000 USD	%	Expenses	'000 USD	%
Project funds	355	79	Personnel	302	76
Core funds	70	16	Office rental	35	9
Data web services	19	4	Network registration	15	4
Donations	5	1	Travel	14	4
			Supplies & services	13	3
			Others (forex gain/loss, bank charges, depreciation, representation)	20	4
TOTAL	449	100	TOTAL	399	100

Income from short-term projects comprised the bulk (79%) of Q-quatics revenues, followed by core funding at 16%, data and web services at 4%, and donations at 1%. We received core funding support from personal support from Dr Daniel Pauly (fifth year in a row); institutional support from CSIRO (third annual support since 2021); Blue Abacus; and Bloom Association.

As normal in a knowledge organization, the bulk of expenses were spent for personnel constituting 76% of total costs and followed by office rent at the IRRI campus at 9%. Network registration was for internet connectivity and IT services for 26 computers. Travel costs included trips for projects, networking, and operational activities.

Staff Assessments and Inflationary Adjustments

Q-quatics implemented the yearly automatic one-step salary increase for the concerned staff who were in Job Grades 1-4, progressing from entry-level to mid-level, and were assessed as performing satisfactorily in 2022. Effective April 2023, five staff received their salary adjustments. In addition, there were also salary adjustments implemented for eight staff who were evaluated as performing satisfactorily and with increased responsibilities. The high inflation rates brought about by the Covid-19 outbreak in 2019 and the recent global developments (e.g., Russian-Ukraine war, Israeli-Palestinian conflict) were also felt in the Philippines. The annual inflation rates increased from 2.4% in 2020 to 6.0% in 2023. To cushion the impact of the lower purchasing power of the Philippine peso, the Board approved a lumpsum inflationary relief in 2023.

Challenges

As in the past, the operations of Q-quatics were largely dependent on the income received from projects. We aspire to an increase in core funding to have greater capacity for the strategic research direction and services needed by our database users and enlarge the user base. We aim at a working annual budget of 600,000 USD which would allow more innovation and, importantly, cover inflationary loss of purchasing power.

Finding a solution to stabilize and increase our financial capabilities remained a challenge throughout the year. We will continue to approach and pitch Q-quatics expertise and products to investors to support our work which provides global knowledge and information systems as a public good necessary for fisheries management, climate change mitigation, and a range of other actual and potential applications. Q-quatics offers an invaluable resource towards initiatives to protect aquatic resources, biodiversity, and our environment for us and future generations.

Our immediate, specific goals are to seek continued excellence in service for current users and partners and reach into new user groups; recruit a new Executive Director; enlarge the membership of the Board of Trustees with highly respected individuals coming from media and/or foundations promoting environmental conservation and social justice; and develop greater capacity in fund raising and catering for funding partner relationships.

Acronyms

AUTH	Aristotle University of Thessaloniki
BRUVS	Baited Remote Underwater Video Systems
EDF	Environmental Defense Fund
CSIRO	Commonwealth Scientific and Industrial Research Organisation
DigIn	Documenting Marine Biodiversity through Digitization of Invertebrate Collections
EEZ	Exclusive Economic Zone
EcoScope	Ecocentric Management for Sustainable Fisheries and Healthy Marine Ecosystems
EPI	Environmental Performance Index
EU	European Union
FB	FishBase
FBC	FishBase Consortium
FFI	Fair-Fish International
FMNH	Florida Museum of Natural History
GEOMAR	Helmholtz Centre for Ocean Research Kiel
HIFMB	Helmholtz Institute for Functional Marine Biodiversity
IT	Information Technology
MAFWR	Ministry of Agriculture, Fisheries, Wealth and Water Resources
MBA	Monterey Bay Aquarium
MNHN	Muséum national d'Histoire naturelle
NSF	U.S. National Science Foundation
NRM	Swedish Museum of Natural History
PIMS	Perry Institute for Marine Science
RMCA	Royal Museum for Central Africa
RMTI	Regional Marine Trophic Index
SWIMS	The Swire Institute of Marine Science
SAU	<i>The Sea Around Us</i>
SLB	SeaLifeBase
UoE	University of Exeter
UL	University of Lorraine
UM	University of Miami
U-M	University of Michigan
UMN	University of Minnesota
UWA	University of Western Australia

Appendix 1. Selected Feedback Received from FishBase and SeaLifeBase Users

Source	Remarks
Ray Simpson @WATtFish via X February 11	I'll miss old Fishbase going back to the early 00's where it was the first time I saw pictures and range information for soooo many cool fishes I'd never heard of for the first time. Excited to see how it looks now and the update was long overdue.
Katie Matthews @katieocean.bsky.social @katie_at_sea via X April 10	FishBase is an unsung hero in ocean research.
Chiara Benvenuto @cbenvenuto73 via X May 28	Did I mention that, among the other things, I also had the huge honour to meet Rainer Froese, the co-founder of @FishBaseProject!!!! Such an honour! I am so grateful for FishBase! @chrysagubili @RominaSSBatista
@MAFWR_FishRes (المديرية العامة للبحوث السمكية - سلطنة عمان) via X July 18	As part of efforts to improve the Sultanate of Oman's classification in the fisheries index within the global Environmental Performance Index (EPI), the Ministry, represented by the General Directorate of Fisheries Research, signed a cooperation agreement with the Q-quatics

Source	Remarks
	organization to update the Sultanate's data in global fish databases. [Google translation from Arabic] @SeaAroundUs @fishbase Sealifebase
R. KellerKopf @KellerKopf via X October 4	Even if you're not interested in fish, the volume and scope of data on 30,000+ freshwater and marine fish species, curated by FishBase, is seriously worth checking out. Simply, an amazing resource.
EmilyTroyer @Fish_Fetisher via X Sept 11	Love FishBase
Cheuk-Ho Wu @SWIMSHKU FishBase Symposium via X September 4	#FishBase is supporting lots of work that promote #oceanliteracy . We are happy to be part of FishBase in helping more people understand the #oceans ."
Leen Vandepitte Flanders Marine Institute, Belgium FishBase Symposium via X September 4	FishBase is out there. #FishBase is the way. We will further consolidate collaboration with FishBase and SeaLifeBase
WorldFish @WorldFishCenter via X/Twitter November 14	Honored and humbled to be a part of this revolutionizing resource's legacy tracing back to 1990 when it started out as a collaboration between @WorldFishCenter and @FAOfish and other partners. A

Source	Remarks
	big shoutout to all who contributed to its success!
Reuben Goforth, PhD Associate Professor of Aquatic Ecology, Department of Forestry and Natural Resources, Purdue University, September	I have to say that I am impressed with FB's management of all this information! It seems like keeping up with changes in names, distributions, etc., is extremely challenging (more so than I expected). Thank you very much for all your efforts!
Giornale di Sicilia Gds.it @GDS it via X December 8	FishBase, the most complete online global database of information on known fish species, is enriched with new information on the Sicilian macrostigma trout. [Translated from Italian]

Appendix 2. Publications by Q-quatics Members as Co-authors and by Project Partners

Abucay L, Sorongon-Yap P, Kesner-Reyes K, Capuli EC, Reyes RJr, Daskalaki E, Ferra C, Scarcella G, Coro G, Ordines F, Sánchez-Zulueta P, Dakalov G, Klayn S, Celie L, Scotti M, Gremilet D, Lambert C, Gal G, Palomares MLD, Dimarchopoulou D, Tsikliras AC (2023) Scientific knowledge gaps on the biology of non-fish marine species across European Seas. *Frontiers in Marine Science* 10:1198137.

Atwood TB, Sala E, Mayorga J, Bradley D, Cabral RB, Auber A, Cheung W, Ferreti F, Friedland AM, Gaines SD, Garilao C, Goodell W, Halpern BS, Hinson A, Kachner K, Kesner-Reyes K, Lepriur F, McGowan J, Morgan LE, Mouillot D, Palacios-Abrantes J, Possingham HP, Rechberger KD, Worm B, Luncheon J (2023) Reply to: Quantifying the carbon benefits of ending bottom trawling. *Nature* 617:E3-E5.

Chardard D, Willig C, Bailly N, Atanacio R, Fordoxcel L, Delaunay S, Gerard PA, Teletchea F (2023) The onset of a collaboration between FishBase and aquariums: the example of the Muséum-Aquarium of Nancy. *Cybio* 47(3):213-224.

Colchen T, Ledoré Y, Fontaine P, Teletchea F, Pasquet A (2023) Larval pikeperch *Sander lucioperca* cannibals are more efficient predators on zebrafish *Danio rerio* than non-cannibals. *Aquaculture* 575:739756.

Coro G, Bove P, Kesner-Reyes K (2023) Global-scale parameters for ecological models. *Scientific Data* 10(7):1-21.

Davis A, Pauly D (2023) Kenneth Richard Ruddle (1942-2023). *Maritime Studies* 22(4):43

- Demirel N, Nauen CE, Palomares MLD (2023) Fishing effort and the evolving nature of its efficiency. *Frontiers in Marine Science* 10:1180174.
- Derrick B, Burns K, Zhu A, Andreoli V, Zeller D, Pauly D (2023) Small-scale fisheries catch and fishing effort in the Socotra Archipelago (Yemen) between 1950 and 2019. *Frontiers in Marine Science* 10:1201661.
- Fredston AL, Cheung WWL, Frölicher TL, Kitchel ZJ, Maureaud AA, Thorson JT, Auber A, Merigot B, Palacios-Abrantes J, Palomares MLD, Pecuchet L, Shackell NL, Pinsky ML (2023) Marine heatwaves are not dominant driver of change in demersal fishes. *Nature* 621(7978):324-329.
- Froese R, Coro G, Palomares MLD, Bailly N, Scotti M, Froese T, Garilao C, Pauly D (2023) A simple framework for the exploration of functional biodiversity. *Cybio* 47(3):271-286.
- Froese R, Pauly D (2023) Comment on "Metabolic scaling is the product of life-history optimization". *Science* 380(6643):eade6084.
- Froese R, Winker H, Coro G, Palomares MLD, Tsikliras AC, Dimarchopoulou D, Touloumis K, Demirel N, Vianna G, Scarcella G, Schijns R, Liang C, Pauly D (2023) New developments in the analysis of catch time series as the basis for fish stock assessments: The CMSY++ method. *Acta Ichthyologica et Piscatoria* 53:173-189.
- Gissi E, McDevitt-Irwin JM, Kachner K, Kesner-Reyes K, Hazen E, Santoleri R, Micheli F (2023) Identifying climate refugia through analysis of functional diversity. *Research Square*. <https://doi.org/10.21203/rs.3.rs-3554317/v1>.

- Heidrich KN, Meeuwig JJ, Juan-Jorda MJ, Palomares MLD, Pauly D, Thompson CDH, Friedlander AM, Sala E, Zeller D (2023) Multiple lines of evidence highlight the dire straits of yellowfin tuna in the Indian Ocean. *Ocean and Coastal Management* 246:106902.
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- Khalfallah M, Mahmoud HH, Fahim RM, Pauly D (2023) Once upon a century, the Egyptian Mediterranean fisheries (1920–2019), as affected by ‘fishing down’ and climate change. *Ocean and Coastal Management* 245(1):106831.
- Lavin CP, Pauly D, Dimarchopoulou D, Liang C, Costello MJ (2023) Fishery catch is affected by geographic expansion, fishing down food webs and climate change in Aotearoa, New Zealand. *PeerJ* 11:e16070.

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- Nauen CE (2023) Art of hosting approaches with greater participation of scientists can support robust solutions for increased societal resilience. In *EGU General Assembly Conference Abstracts*, pp. EGU-4421.
- Nauen CE, Rius-Barile J, Palomares MLD (2023) Know your fish – venturing further with the FishBase Guide app. *Cybio* 47(3):309-313.

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Appendix 3. Q-quatics Board of Trustees

Nauen, Cornelia Elisabeth

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Philippines

Appendix 4. Q-quatics List of Staff and Directors

Directors

Bimbao, Mary Ann P.	Executive Director
Palomares, Maria Lourdes D.	Science Director
Bailly, Nicolas	Taxonomy Director

Research Group

Abucay, Luisa R.	Research Analyst
Bato-David, Elizabeth A.	Research Analyst
Capuli, Estelita Emily D.	Taxonomy Group Lead
Hernandez, Roesyvette V.	Research Assistant*
Jansalin, Jeremiah Glenn M.	Research Assistant
Maquinano, Joan Kate R.	Research Assistant**
Mendoza, Jasper Brian M.	Research Assistant
Parakikay, Timothy P.	Research Assistant
Parducho, Vina Angelica A.	Research Analyst
Reyes, Kathleen K.	Research Associate
Reyes, Rodolfo Jr. B.	Research Associate
Taguines, Angele Paulee V.	Research Assistant***
Yap, Patricia Marjorie S.	Research Associate

Information Technology Group

Agbay, Vanessa Camille A.	Software Engineer
Barile, Ma Josephine France R.	IT Group Lead
Nevado, Martin Paulo D.	Software Engineer
Ortiz, Shelumiel	Software Engineer

Administration and Finance

Borbe, Cynthia R.	Finance Officer
Luna, Susan M.	Executive Assistant/Interim
	Traits Group Lead
Painaga, Geezle Joy V.	Administrative Assistant

Common Services

Atanacio, Rachel C.	Artist Consultant
Banasihan, Lyra Joyce P.	Librarian
Glorioso, Joann R.	Communication Officer
Lagasca, Frederick C.	Computer Maintenance
	Technician
Yap, Michael Angelo A.	Artist

*Resigned 31 October/Started 1 November

**Resigned 30 September

***Started 01 December

Appendix 5. Q-quatics Projects, Data Services, and Core Funding

Title	Donor/ Partner	Project Duration	Status
Projects			
Understanding the role of aquatic foods in helping to alleviate food security and nutrition vulnerabilities especially in the face of climate change	EDF, via SAU	10 Sept 2021 – 15 Mar 2023	Completed, from an ongoing project
Fish micronutrients as a measure of fisheries catch in <i>Sea Around Us</i> database	Oceana, via SAU	14 Feb 2022 – 30 Jun 2023	Completed, from an ongoing project
Extension of the freshwater fisheries project with UMN	UMN, via SAU	1 Jan 2023 – 31 Dec 2023	Completed, new project in 2023
New species and species updates in FishBase	NRM	30 Jan 2023 – 31 Mar 2023	Completed, new project in 2023
FishBase data and web services 2023	GEOMAR	1 Feb 2023 – 31 Dec 2023	Completed, new project in 2023
FishBase Guide App, Ver2 (for Android phones)	<i>Mundus maris</i>	1 May 2023 – 1 Jun 2023	Completed, new project in 2023
Data and web services 2023	HIFMB	1 May 2023 – 31 Dec 2023	Completed, new project in 2023

Title	Donor/ Partner	Project Duration	Status
Ontology and FishBase Taxonomy	WorldFish	1 Sep 2023 – 15 Dec 2023	Completed, new project in 2023
Development and updating of the Sultanate of Oman's data in global databases (SAU, FB, SLB) with the aim of improving the country's ranking in the EPI	MAFWR, via SAU	12 Jul 2023 – 11 Jan 2024	New project in 2023, ending in 2024
Bahamas Fish Stock Assessment	PIMS, via SAU	8 Nov 2023 – 8 May 2024	New project in 2023, ending in 2024
Atlas of Fish Species in France	MNHM	1 Dec 2023 – 31 Dec 2024	New project in 2023, ending in 2024
SWIMS-FB/SLB Collaboration on International Marine Databases (P3)	SWIMS, via SAU	1 Jan 2023 – 31 Dec 2025	New project in 2023, ending in 2025
Analysis of biodiversity of Saint-Pierre and Miquelon; Update of list of species in Saint-Pierre and Miquelon in FishBase and SeaLifeBase	UL, via SAU	Oct 2023 – Dec 2025	New project in 2023, ending in 2025
Documenting Marine Biodiversity through Digitization	NSF, thru FMNH	30 Nov 2022 – 14 Nov 2024	Ongoing project, ending in 2024

Title	Donor/ Partner	Project Duration	Status
of Invertebrate Collections (DigIn)			
FishBase Data Platform Extension for BRUVS	Minderoo Foundation, via UWA	31 Aug 2020 – 31 Jul 2025	Ongoing project, ending in 2025
Ecocentric Management for Sustainable Fisheries and Healthy Marine Ecosystems (EcoScope)	EU H2020-BG-2020-2/24 Institutions majority from Europe, AUTH as Coordinator	1 Sept 2021 – 31 Aug 2025	Ongoing project, ending in 2025
Provision of cell-based wild-captured seafood datasets	MBA, via SAU	Oct 2023 – Sept 2026	New project in 2023, ending in 2026
Convex Seascape Survey	UoE, via SAU	1 Jul 2022 – 31 Mar 2027 (27 Jan 23):QQ	New project in 2023, ending in 2027
IT Services for SAU			
IT services	Oak Foundation, via SAU	1 Apr 2019 – 31 Mar 2023	Completed
Data/Web Services			
Global spatial data for all years (1950 - 2019), gears and taxa	McKinsey & Company, via SAU	18 May 2023	Completed, one-time request
SAU Data request (Global EEZ shapefile)	Marvivo.earth, via SAU	20 Jun 2023	Completed, one-time request

Title	Donor/ Partner	Project Duration	Status
Global cell data by fishing entity, taxon name, commercial group, year and catch in 19 years	U-M, via SAU	12 Jul 2023	Completed, one-time request
Updated stock assessment spreadsheet as of December 2023	NORMA	Dec 2023	Completed, with possibility of continuing service
Stock Assessment Updates	FFI	yearly, since 2018	Continuous
Updated stocks spreadsheet	Shifting Values e.U.	yearly, since 2020	Continuous
Core Funding			
Contributions to FishBase/SeaLife-Base	Blue Abacus	May 2023	With possibility of continuing support
General support to FishBase/SeaLife-Base	Bloom Association	Dec 2023	With possibility of continuing support
Personal donation	Daniel Pauly	Yearly since 2019	With possibility of continuing support
Support to FishBase/SeaLife-Base	CSIRO	1 Jul 2022 – 30 Jun 2023	Continuous since July 2020
Pipeline			
Fish ID for Timor Leste Species	WorldFish	1 Jan 2023 – 31 Jan 2024	New project in 2024,

Title	Donor/ Partner	Project Duration	Status
			ending in 2024
On AquaMaps	UM	2024	Pledge support
On BRUVS	WorldFish	2024	Proposal in preparation



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Quantitative Aquatics

Khush Hall, IRRI, College, Los Baños,
Laguna, Philippines